Supplement to STAR methods section
Kinetic modeling of RNA flow:
ordinary differential equations (ODEs) and solutions

- Whole Cell (Total RNA) turnover rate k_{WC}
  ODE: $\frac{dT(t)}{dt} = k_p - k_{WC}T(t)$
  Fraction new RNA solution: $\Lambda_{WC}(k_{WC}, t) = 1 - \exp(-k_{WC}t)$
- Chromatin turnover rate k_{CH}
  ODE: $\frac{dCH(t)}{dt} = k_p - k_{CH}CH(t)$
  Fraction new RNA solution: $\Lambda_{CH}(k_{CH}, t) = 1 - \exp(-k_{CH}t)$
- Nuclear turnover rate k_N
  ODE: $\frac{dN(t)}{dt} = k_p - k_N N(t)$
  Fraction new RNA solution: $\Lambda_N(k_N, t) = 1 - \exp(-k_N t)$
- Cytoplasmic turnover rate k_{CY}
  ODE: $\frac{dCY(t)}{dt} = k_N N(t) - k_{CY}CY(t)$
  Fraction new RNA solution: $\Lambda_{CY}(k_N, k_{CY}, t) = \frac{k_{CY}}{k_{CY}-k_N}(1 - \exp(-k_N t)) + \frac{-k_N}{k_{CY}-k_N}(1 - \exp(-k_{CY}t))$
- Polysome loading rate k_{PL}
  ODEs (for Mature unbound and Polysome bound RNA): $CY(t) = M(t) + P(t) \Longrightarrow$
  $\frac{dM(t)}{dt} = k_N N(t) - (k_{CY} + k_{PL})M(t)$
  $\frac{dP(t)}{dt} = k_{PL}M(t) - k_{CY}P(t)$
  Fraction new RNA solution:

$$\Lambda_P(k_N, k_{CY}, k_{PL}, t) =$$

$$1+\frac{-k_{CY}(k_{CY}+k_{PL})}{(k_{CY}+k_{PL}-k_N)(k_{CY}-k_N)}\exp(-k_N t)+\frac{-k_N k_{CY}}{k_{PL}(k_{CY}+k_{PL}-k_N)}\exp(-(k_{CY}+k_{PL})t)+\frac{k_N(k_{CY}+k_{PL})}{k_{PL}(k_{CY}-k_N)}\exp(-k_{CY}t)$$

For non-PUNDs: genes without evidence for nuclear RNA degradation

- Chromatin release rate $k_R = k_{CH}$
- Nuclear export rate k_E
  ODE: $\frac{dN(t)}{dt} = k_p - k_E(N(t) - CH(t))$
  Fraction new RNA solution: see solution of nuclear export rate for PUNDs below, with nuclear degradation rate k_{ND} set to 0.

For PUNDs: genes Predicted to Undergo Nuclear RNA Degradation

- Nuclear degradation k_{ND}
  ODE (for whole cell, Total RNA): $T(t) = N(t) + CY(t) \Longrightarrow$
  $\frac{dT(t)}{dt} = k_p - k_{ND}N(t) - k_{CY}CY(t)$
  Fraction new RNA solution:
  $\Lambda_T(k_R, k_E, k_{ND}, k_{CY}, t) = 1 + \Theta[\Psi\exp(-at) + \Phi\exp(-bt) + \Omega\exp(-k_{CY}t)]$
  with
  $a := k_{CH} = k_R + k_{ND}$
  $b := k_E + k_{ND}$
  $\Theta := \frac{bk_{CY}}{k_{CY}(a+b-k_{ND})+(a-k_{ND})(b-k_{ND})}$
  $\Psi := \frac{(b-k_{ND})(k_{CY}-k_{ND})}{(a-b)(k_{CY}-a)}$
  $\Phi := \frac{-a(a-k_{ND})(k_{CY}-k_{ND})}{b(a-b)(kD-b)}$
  $\Omega := \frac{-a(a-k_{ND})(b-k_{ND})}{k_{CY}(k_{CY}-a)(k_{CY}-b)}$

- Chromatin release rate $k_R = k_{CH} - k_{ND}$

- Nuclear export rate k_E:
  ODE (for Nuclear RNA): $\frac{dN(t)}{dt} = k_p - k_{ND}N(t) - k_E(N(t) - CH(t))$
  Fraction new RNA solution:

$$\Lambda_N(k_R, k_E, k_{ND}, t) = 1 + \frac{b\,(b - k_{ND})\exp(-at) - a\,(a - k_{ND})\exp(-bt)}{(a + b - k_{ND})\,(a - b)}$$

  with a and b as defined above.